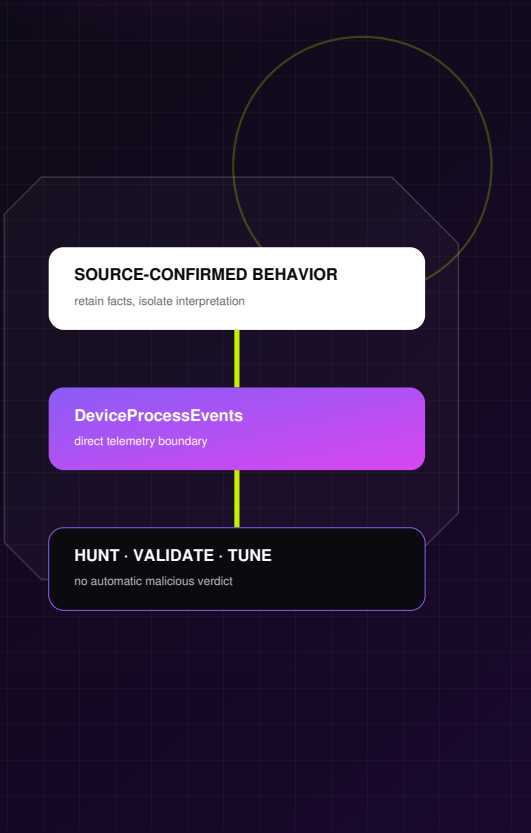


GUNRA RANSOMWARE WMIC RECOVERY INHIBITION

Behavioral candidate for source-observed volume shadow-copy deletion through WMIC before ransomware impact.

PRODUCTION CANDIDATE

CONFIDENCE: MEDIUM



REPORT DATE
11 August 2026

VERSION
v1.0

PRIMARY TELEMETRY
Microsoft Defender XDR ·
DeviceProcessEvents

ATT&CK
T1490

Evidence-led dossier. Query status remains an untested implementation sketch until executed in the target environment.
INDEPENDENT RESEARCH FORMAT · CONTROLLED UI/UX MASTER



EXECUTIVE ASSESSMENT

Decision-grade summary for detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

Detect the destructive recovery-inhibition command, not a Gunra hash.

The joint FBI/CISA/NSA advisory documents Gunra actors invoking WMIC.exe with shadowcopy and delete before encryption. The complete command shape is observable in process telemetry and remains relevant beyond one ransomware family.

OPERATIONAL VALUE

High-signal pre-impact behavior with direct triage context.

The logic identifies an action that removes a principal Windows recovery mechanism. It uses no campaign IOC, threshold, rarity, aggregation or inferred outcome and returns the initiating process, account, remote-session context and hashes for immediate investigation.

9.2

01

OPERATIONAL VALUE



Medium

02

DETECTION CONFIDENCE



Candidate

03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACTS

Observed behavior

- AA26-222A reports Gunra deleting volume shadow copies through WMI before encryption.
- The advisory provides an example command using WMIC shadowcopy where ID equals a GUID delete.
- Gunra also deleted backup and archived data in one victim environment.

ANALYTICAL ASSESSMENT

What the evidence supports

- WMIC-based shadow-copy deletion is reusable recovery-inhibition behavior, not only a Gunra artifact.
- Requiring executable, object class and delete verb is materially more selective than alerting on WMIC alone.
- A match requires authorization review and does not independently prove ransomware execution.

DETECTION ASSUMPTIONS

What must be validated

- MDE records complete process command lines on covered Windows endpoints.
- The tenant does not routinely delete shadow copies through this exact command outside controlled operations.
- Initiating-process and account context are sufficiently populated for triage.

Confidence boundary

The command can be used legitimately by backup, recovery, imaging or administrative workflows. The query was not executed in the target tenant, so it remains a Medium-confidence candidate rather than production-ready detection.



THREAT MECHANICS

Source-observed chain and explicit observability boundaries.

ATTACK FLOW

01

Initial access

Affiliates exploit exposed systems or valid credentials.

02

Privilege and movement

Impacket, RDP and SSH support internal access.

03

Data collection

Sensitive data is archived and exfiltrated.

04

Recovery inhibition

WMIC deletes selected volume shadow copies.

05

Encryption impact

Gunra encrypts eligible data and writes ransom notes.

Observable core

A WMIC process whose command line contains both the shadowcopy object class and the delete verb, with device, account, process lineage and remote-session context.

Non-observable boundary

Whether deletion succeeded, the target GUID, backup coverage, subsequent encryption, data exfiltration and Gunra attribution.

DETECTION HYPOTHESIS

Recovery inhibition should be visible before or during ransomware impact.

If an attacker uses the documented WMIC method to remove Windows volume shadow copies, DeviceProcessEvents should record WMIC.exe with shadowcopy and delete in ProcessCommandLine. Legitimate activity can be distinguished through approved change records, backup tooling, operator identity, device role and surrounding destructive or ransomware behavior.

REQUIRED TELEMETRY

DeviceProcessEvents

- Microsoft Defender for Endpoint process creation events.
- FileName and complete ProcessCommandLine.
- Device, process and event identifiers.

OPTIONAL ENRICHMENT

Context that improves disposition

- Initiating process, user and elevation context.
- Remote-session source and adjacent logons.
- Backup-control events and later file-impact telemetry.

UNAVAILABLE HERE

Do not infer

- WMIC return code or confirmed deletion result.
- Shadow-copy inventory before and after execution.
- Ransomware payload identity from this event alone.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and quality gates for a defensible hunt.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Defender XDR - DeviceProcessEvents

DeviceProcessEvents documents process creation with executable, command line, user, elevation, lineage, remote-session, hash, device and unique-event context.

EVIDENCE BOUNDARY

A matching event is a lead, not a verdict.

The table proves command invocation. It does not prove successful deletion, loss of recoverability, encryption or attribution.

REQUIRED AND ENRICHMENT FIELD MODEL

CATEGORY	FIELD	PURPOSE	REQUIREMENT
Process event	DeviceProcessEvents	Primary command invocation.	REQUIRED
Executable	FileName, FolderPath	WMIC image and location.	REQUIRED
Command	ProcessCommandLine	shadowcopy and delete evidence.	REQUIRED
Device	DeviceId, DeviceName	Endpoint pivot.	REQUIRED
Process key	ProcessUniqueId, ProcessId	Timeline correlation.	REQUIRED
Identity	AccountName, AccountUpn	Authorization review.	REQUIRED
Lineage	InitiatingProcessFileName, InitiatingProcessCommandLi ne	Origin context.	REQUIRED
Session	IsProcessRemoteSession, ProcessRemoteSessionIP	Remote-use enrichment.	OPTIONAL
Event identity	Timestamp, ReportId	Documented event tuple.	REQUIRED

Data quality requirements

- Confirm MDE coverage and full command-line retention.
- Execute a harmless lab validation against a disposable shadow copy.
- Review all matches across native retention.
- Map approved backup and imaging workflows before enabling alerting.

Known failure conditions

- Process telemetry is absent or command line is truncated.
- The actor uses PowerShell, vssadmin, diskshadow or direct APIs instead.
- WMIC is renamed or proxied through another execution path.
- Legitimate automation reproduces the full pattern without stable ownership context.



DETECTION LOGIC

Primary-platform logic with every condition traceable to evidence.

PRODUCTION CANDIDATE

DETECTION OBJECTIVE

Find source-observed WMIC recovery inhibition.

Select WMIC executions containing the shadowcopy object class and delete verb, then return the evidence needed for authorization and impact review.

ANALYTICAL LOGIC

Direct filters and investigable output.

Three direct predicates; no join, sequence, threshold, aggregation, rarity, IOC, wildcard filename or automatic Gunra attribution.

KQL · UNTESTED CANDIDATE

```
// Gunra / WMIC shadow-copy deletion
// Classification: Production candidate - Untested implementation sketch
// Confidence: Medium
// Primary platform: Microsoft Defender XDR Advanced Hunting
// Required telemetry: Microsoft Defender for Endpoint process creation events
// Source evidence: AA26-222A documents the exact WMIC shadowcopy delete behavior.
// No threshold, rarity, join, sequence, hash, domain, or campaign attribution is used.

DeviceProcessEvents
| where FileName =~ "WMIC.exe"
| where ProcessCommandLine has "shadowcopy"
| where ProcessCommandLine has "delete"
| project
    Timestamp,
    DeviceId,
    DeviceName,
    AccountName,
    AccountUpn,
    FileName,
    FolderPath,
    ProcessCommandLine,
    ProcessId,
    ProcessUniqueId,
    ProcessIntegrityLevel,
    ProcessTokenElevation,
    InitiatingProcessFileName,
    InitiatingProcessCommandLine,
    InitiatingProcessAccountName,
    InitiatingProcessAccountUpn,
    IsProcessRemoteSession,
    ProcessRemoteSessionIP,
    SHA1,
    InitiatingProcessSHA1,
    ReportId
| order by Timestamp desc
```

STATUS	THRESHOLD	JOIN / SEQUENCE	PRIMARY KEY	DISPOSITION
Untested candidate	None	None	DeviceName + ProcessUniqueId + Timestamp	Manual triage

What a match means

An endpoint invoked the complete WMIC shadow-copy deletion pattern and requires urgent validation of authorization, execution result and surrounding impact.

What a match does not prove

Successful deletion, loss of backups, file encryption, data theft, malicious intent or Gunra attribution.



TRIAGE & INVESTIGATION

Evidence collection before escalation.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the raw process event.
Verify executable path, full command line, device, account and process key.
- 02

Establish authorization.
Check backup, recovery, imaging, maintenance and change records.
- 03

Review execution lineage.
Inspect parent process, elevation, remote session and neighboring logons.
- 04

Assess recovery state.
Verify shadow-copy and backup status through endpoint or backup-platform evidence.
- 05

Scope impact.
Search for archive creation, exfiltration tools, mass file modification and ransom notes.

Escalation gate

Escalate when the command is unauthorized, remotely initiated, executed from suspicious lineage, followed by missing recovery points or adjacent collection, exfiltration or encryption evidence.

Primary benign overlap

Backup products, disaster-recovery testing, imaging, storage maintenance and administrators may intentionally remove snapshots. Tune only with verified owner, tool, device role and change scope; never allowlist WMIC globally.

INVESTIGATION PIVOTS

Host
Role, coverage and business criticality.

Identity
Operator, privilege and session history.

Parent
Shell, RMM, Impacket or management lineage.

Recovery
VSS inventory and backup health.

Files
Mass modification and ransom-note creation.

Network
Remote session and exfiltration activity.

DISPOSITION CRITERIA

OUTCOME	MINIMUM EVIDENCE	ACTION
BENIGN	Approved, owned and repeatable activity.	Document and tune narrowly.
SUSPICIOUS	Unknown activity without corroborated malicious follow-on.	Deepen endpoint review.
LIKELY MALICIOUS	Unauthorized match plus coherent file, process, network or control-plane evidence.	Escalate incident response.



FALSE POSITIVES & VALIDATION

Operational controls required before promotion.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	WHY IT OVERLAPS	TUNING ACTION
Backup or recovery maintenance	May remove obsolete shadow copies.	Require approved product, service account, host group and change record.
Imaging or lab reset	Disposable systems may clear recovery state.	Scope to managed lab assets and maintenance windows.
Administrator troubleshooting	Manual cleanup can reproduce the command.	Validate operator, ticket, target GUID and necessity.

Baseline plan

- Search 30-90 days or available retention.
- Inspect every match and identify owners.
- Measure command-line and lineage completeness.
- Do not introduce a count or rarity threshold.

Validation procedure

- Create a disposable shadow copy in an isolated host.
- Run the exact documented WMIC method and confirm the event.
- Verify non-destructive WMIC queries do not match.
- Review historical matches with backup and Windows platform owners.

TEST CASES

CASE	EXPECTED RESULT
Controlled WMIC deletion in an isolated Windows lab	Expected match with complete command and lineage.
WMIC query of shadowcopy without delete	No match.
Unrelated WMIC delete against another class	No match because shadowcopy is absent.
vssadmin or PowerShell deletion	Expected miss; document adjacent coverage requirement.

Hunting Initial query	Candidate Current status	Production Measured precision	Optimized Stable feedback
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Success

The lab command matches, benign queries do not, fields are complete and historical overlap is explainable through durable ownership context.

Rejection

Command lines are incomplete or legitimate activity frequently reproduces the pattern without a reliable discriminator.

Failure

The controlled command produces no usable DeviceProcessEvents record.



ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

MITRE ATT&CK MAPPING

TECHNIQUE	NAME	COVERAGE STATEMENT
T1490	Inhibit System Recovery	Directly covers deletion of Windows volume shadow copies to impair recovery.

Detection coverage summary

Strong for WMIC commands containing shadowcopy and delete; weak for alternative utilities, direct APIs, renamed tooling or uncollected process events.

Residual risk

Attackers can use vssadmin, PowerShell, diskshadow, WMI APIs or backup-platform deletion, while successful loss of recovery requires separate control-plane evidence.

ANALYST CONCLUSION

A defensible behavioral candidate with a clear validation burden.

The government advisory directly documents the behavior and Microsoft documents every required field. The command combination is specific enough for a Production candidate, but tenant execution, benign-overlap measurement and recovery-state validation remain mandatory before alerting.

REFERENCES

- 1. FBI, CISA, DC3, NSA, USSS and KNPA, “#StopRansomware: Gunra Ransomware,” 10 August 2026; Gunra observed from April 2025 and RaaS expansion from January 2026. [https://media\[.\]defense\[.\]gov/2026/Aug/10/2003976697/-1/-1/0/CSA_STOPRANSOMWARE_GUNRA_RANSOMWARE.PDF](https://media.defense.gov/2026/Aug/10/2003976697/-1/-1/0/CSA_STOPRANSOMWARE_GUNRA_RANSOMWARE.PDF)
- 2. Microsoft Learn, “DeviceProcessEvents table,” updated 7 August 2026; accessed 11 August 2026. [https://learn\[.\]microsoft\[.\]com/en-us/defender-xdr/advanced-hunting-deviceprocessevents-table](https://learn.microsoft.com/en-us/defender-xdr/advanced-hunting-deviceprocessevents-table)
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DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v1.0	11 August 2026	Candidate	Detection Engineering

Publication control

Eight A4 pages. URLs are defanged. No customer data, credentials, HTML source or rendering assets are included in the public repository.